If you have understood this far, let us approach this in another way.

The governing equation of a second order dynamic mechanical system is

$m\ddot{x} + b\dot{x} + kx = F(t).$

Taking Laplace transform on both sides

$m\{s^2X(s) - sx(0) - \dot{x}(0)\} + b\{sX(s) - x(0)\} + kX(s) = F(s)$

$(ms^2 + bs + k)X(s) - \{(ms + b)x(0) + m\dot{x}(0)\} = F(s)$ ....(15)

$(ms^2 + bs + k)X(s) = \{(ms + b)x(0) + m\dot{x}(0)\} + F(s)$

$$X(s) = \frac{\{(ms+b)x(0)+m\dot{x}(0)\}+F(s)}{ms^2+bs+k}$$

$$= \frac{\{(ms+b)x(0)+m\dot{x}(0)\}}{ms^2+bs+k} + \frac{F(s)}{ms^2+bs+k} \text{ ....(16)}$$

The first term on RHS will give the natural response (homogeneous solution) and the second term will give the forced response (particular solution)

If we know the initial conditions and F(s) {which is the Laplace Transform of the input F(t) }, we can find out the response x(t) by taking the inverse Laplace Transform of (16)

Assume that no input acts on the system [F(s)=0] and initial conditions $x(0) = 1, \ and \ \dot{x}(0) = 0$

$$\boldsymbol{X(s) = \frac{\{(ms + b)x(0) + m\dot{x}(0)\}}{ms^2 + bs + k}} \dots\dots\dots\dots\dots\dots\dots\dots\dots\dots.(17)$$

$$= \frac{\{(ms+b).1+m.0\}}{ms^2+bs+k} = \frac{s+\frac{b}{m}}{s^2+\frac{b}{m}s+\frac{k}{m}} = \frac{s+2\zeta\omega_n}{s^2+2\zeta\omega_n s+\omega_n{}^2}$$ (Refer Eq.(8) and (9))

x(t) can be calculated by taking the invers transform. Let us write X(s) as partial fraction

$$X(s) = \frac{s+2\zeta\omega_n}{(s+\zeta\omega_n)^2-(\zeta\omega_n)^2+\omega_n{}^2} = \frac{s+2\zeta\omega_n}{[s+\zeta\omega_n]^2-[\omega_n\sqrt{\zeta^2-1}]^2} \text{ .... } \mathbf{(18)}$$

Let us assume that the system is critically damped ($\zeta = 1$), The equation (18) reduces to

$$X(s) = \frac{s + 2\omega_n}{(s + \omega_n)^2} = \frac{A}{s + \omega_n} + \frac{B}{(s + \omega_n)^2} \qquad (Partial\ Fraction)$$

$s + 2\omega_n = A(s + \omega_n) + B$...**(19a)**

By letting $s = -\omega_n$, we get B= ω_n

Equating constants on both sides of (19a), we can write $2\omega_n = A\omega_n + B$

We get A=1

$$X(s) = \frac{1}{s + \omega_n} + \frac{\omega_n}{(s + \omega_n)^2}$$

Taking inverse Laplace Transform

$x(t) = e^{-\omega_n t} + \omega_n(e^{-\omega_n t})\, t = e^{-\omega_n t}(1 + \omega_n t) \dots\dots\dots\dots\dots\dots\dots.(19b)$

This is exactly same as (12) with A_1=1, A_2=ω_n

(You will be getting the same expression as (19b) when you do #Exercise 5 for the critically damped system with the same initial conditions. Verify!!!!!!)

Let us consider an overdamped system ($\zeta > 1$). The equation (18) is written as (for simplification)

$$X(s) = \frac{s+2a}{(s+a)^2-b^2} \qquad \text{where } a = \zeta\omega_n \text{ and } b = \omega_n\sqrt{\zeta^2-1}$$

$$X(s) = \frac{s+2a}{\{(s+a)+b\}\{(s+a)-b\}} = \frac{A}{s+a+b} + \frac{B}{s+a-b}$$

$s + 2a = A(s+a-b) + B(s+a+b)$. Let us put s=b-a and s=-a-b to evaluate A and B

$$b - a + 2a = A(b-a+a-b) + B(b-a+a+b) \quad \gg \quad B = \frac{a+b}{2b}$$

$$-a - b + 2a = A(-a-b+a-b) + B(-a-b+a+b) \quad \gg \quad A = -\frac{a-b}{2b}$$

$$X(s) = \frac{-\frac{a-b}{2b}}{s+a+b} + \frac{\frac{a+b}{2b}}{s+a-b} = \frac{1}{2b}\left\{\frac{a+b}{s+(a-b)} - \frac{a-b}{s+(a+b)}\right\}$$

Taking inverse Laplace Transform

$$x(t) = \frac{1}{2b}\left\{(a+b)\,e^{-(a-b)t} - (a-b)\,e^{-(a+b)t}\right\}$$

$$x(t) = \frac{1}{2\omega_n\sqrt{\zeta^2-1}}\left\{\left(\zeta\omega_n + \omega_n\sqrt{\zeta^2-1}\right)e^{-\left(\zeta\omega_n-\omega_n\sqrt{\zeta^2-1}\right)t} - \left(\zeta\omega_n - \omega_n\sqrt{\zeta^2-1}\right)e^{-\left(\zeta\omega_n+\omega_n\sqrt{\zeta^2-1}\right)t}\right\}$$

$$x(t) = \frac{e^{-(\zeta\omega_n)t}}{2\omega_n\sqrt{\zeta^2-1}}\left\{\left(\zeta\omega_n + \omega_n\sqrt{\zeta^2-1}\right)e^{\left(\omega_n\sqrt{\zeta^2-1}\right)t} - \left(\zeta\omega_n - \omega_n\sqrt{\zeta^2-1}\right)e^{-\left(\omega_n\sqrt{\zeta^2-1}\right)t}\right\}$$

$$\boldsymbol{x(t) = e^{-\zeta\omega_n t}\left\{\frac{\zeta+\sqrt{\zeta^2-1)}}{2\sqrt{\zeta^2-1}}\,e^{\left(\omega_n\sqrt{\zeta^2-1}\right)t} - \frac{\zeta-\sqrt{\zeta^2-1)}}{2\sqrt{\zeta^2-1}}\,e^{-\left(\omega_n\sqrt{\zeta^2-1}\right)t}\right\}} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\textbf{(20)}$$

Compare this with equation (11) and your #Exercise 5 problem of overdamped system response with same initial conditions. Verify the value of A_1 and A_2 !!!!!

Consider now an underdamped system with $\zeta < 1$, the equation (18) is written as

$$X(s) = \frac{s+2\zeta\omega_n}{[s+\zeta\omega_n]^2-[\omega_n\sqrt{\zeta^2-1}]^2} = \frac{s+2\zeta\omega_n}{[s+\zeta\omega_n]^2-[j\omega_n\sqrt{1-\zeta^2}]^2} = \frac{(s+\zeta\omega_n)+\zeta\omega_n}{[s+\zeta\omega_n]^2+[\omega_d]^2}$$

Recall that $\omega_d = \omega_n\sqrt{1-\zeta^2}$ and j^2=(-1)

$$X(s) = \frac{(s+\zeta\omega_n)}{[s+\zeta\omega_n]^2+[\omega_d]^2} + \frac{\zeta\omega_n}{[s+\zeta\omega_n]^2+[\omega_d]^2} = \frac{(s+\zeta\omega_n)}{[s+\zeta\omega_n]^2+[\omega_d]^2} + \left(\frac{\zeta\omega_n}{\omega_d}\right)\frac{\omega_d}{[s+\zeta\omega_n]^2+[\omega_d]^2}$$

Taking inverse Laplace Transform (revisit the laplace transforms of sin wt and cos wt)

$$x(t) = e^{-\zeta\omega_n t}\cos\omega_d t + \left(\frac{\zeta\omega_n}{\omega_d}\right)e^{-\zeta\omega_n t}\sin\omega_d t \qquad = e^{-\zeta\omega_n t}\left(\cos\omega_d t + \left(\frac{\zeta}{\sqrt{1-\zeta^2}}\right)\sin\omega_d t\right)$$

$$= \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\left(\sqrt{1-\zeta^2}\cos\omega_d t + \zeta\sin\omega_d t\right) = \qquad = \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}}\left(X_0\sin\theta\cos\omega_d t + X_0\cos\theta\sin\omega_d t\right)$$

where $X_0\sin\theta = \sqrt{1-\zeta^2}$ *and* $X_0\cos\theta = \zeta$ so that $X_0^2 = (1-\zeta^2) + \zeta^2 = 1$ and $\tan\theta = \frac{\sqrt{1-\zeta^2}}{\zeta}$

$$x(t) = \frac{1}{\sqrt{1-\zeta^2}}\,e^{-\zeta\omega_n t}\,\sin(\omega_d t + \theta) \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots(21)$$

Compare this with equation (14) and your #Exercise 5 problem of underdamped system response with same initial conditions. Verify the value of X_0 and θ !!!!!

#Exercise 6.

Starting from Equation (17), evaluate the natural responses of over, under, and critically damped second order systems for the following initial conditions. Verify your answer with #Ex. 5. (i) $x(0) = 0,\ \dot{x}(0) = 1$ (ii) $x(0) = 1,\ \dot{x}(0) = 1$

If we set all initial conditions equal to zero, (16) reduces to

$X(s) = \frac{F(s)}{ms^2+bs+k}$.

The transfer function of the system is given by (Recall the definition of Transfer Function)

$$G(s) = \frac{X(s)}{Y(s)} = \frac{1}{ms^2+bs+k} = \frac{1}{m(s^2+\frac{b}{m}s+\frac{k}{m})} = \left(\frac{1}{k}\right)\left(\frac{k}{m}\right)\frac{1}{(s^2+\frac{b}{m}s+\frac{k}{m})} = \left(\frac{1}{k}\right)\frac{\frac{k}{m}}{(s^2+\frac{b}{m}s+\frac{k}{m})}$$

$$G(s) = \left(\frac{1}{k}\right)\frac{{\omega_n}^2}{s^2+2\zeta\omega_n s+{\omega_n}^2}$$

This is now of the form $G(s)=K\frac{p(s)}{q(s)}$.

The standard form of a second order system is considered as $\frac{{\omega_n}^2}{s^2+2\zeta\omega_n s+{\omega_n}^2}$

The characteristic equation is q(s)=0, and the roots of the characteristic equation are called poles of the system

$s^2 + 2\zeta\omega_n s + {\omega_n}^2 = 0$ ……………………………………………………………(22)

$\boldsymbol{s = \frac{-2\zeta\omega_n \pm \sqrt{(2\zeta\omega_n)^2 - 4{\omega_n}^2}}{2}} \quad = \quad \boldsymbol{-\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}}$ ……………(23) [Same as equation (10)]

**Example

The transfer function of a second order system is $G(s) = \frac{25}{s^2+5s+25}$ Find the natural frequency and damping ratio of the system.

Ans: Comparing the given transfer function with the standard form $\frac{{\omega_n}^2}{s^2+2\zeta\omega_n s+{\omega_n}^2}$

${\omega_n}^2 = 25$, and $2\zeta\omega_n = 5$

Hence $\omega_n = 5$ rad/s, and $\zeta = \frac{5}{2\omega_n} = 0.5$

#Exercise 7

Given below are the natural frequency and damping ratio of a second order system respectively. What is system transfer function? Where is the location of the poles?

i) 3 rad/s and 0.25 ii) 5 rad/s and 1 iii) 2 rad/s and 1.25 iv) 4 rad/s and 0

#Exercise 8

Find the natural frequency and damping ratio of the second oerder systems given below

i) $G(s) = \frac{24}{3s^2+8s+24}$ ii) $G(s) = \frac{15}{(s+3)(2s+5)}$

#Exercise 9

The poles of three second order systems are given below. Obtain their transfer functions, damping ratios and natural frequencies.

i) -1, -5 ii) -3,-3 iii) -2±j5